

When Does Corruption Break the Plan?

A Degradation-Tolerance Benchmark for Camera-Only End-to-End Driving

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Abstract

*Camera-only end-to-end (E2E) driving models are nearing deployment, where the camera stream is degraded by blur, noise, low light, weather, frame loss, and memory faults. How much a policy tolerates before its driving breaks is unclear. Corruption-robustness benchmarks target detection or bird’s-eye-view perception, not the planning output that drives the car. We present **DriveDegrade**, a benchmark for image-degradation tolerance in camera-only E2E driving. Sixteen corruption families at five severities are injected on the fly inside the image loader, one operator reaching fifteen policies, and we evaluate open-loop planning on nuScenes and NAVSIM plus a CARLA closed-loop anchor. First, mild degradation barely affects planning, and the families that break it have a clear threshold at mid severity. Second, fragility is corruption-dependent: blur, JPEG, and rain-drop damage planning most, while weather and bit error are tolerated far into the range. Third, a flat curve is ambiguous, so we separate corruptions that degrade the image from those that remove it. A planner that reads its camera must lose accuracy when information is deleted, whatever it does under quality loss. On these two axes the planners separate sharply, quantifying the ego-status shortcut without mistaking indifference for robustness. A released vision-language-action planner is flat on both axes, and blinding all six of its cameras costs it only 11.5%.*

1. Introduction

End-to-end (E2E) camera-only driving has matured quickly: a single network reads the surround-view cameras and predicts the ego trajectory, folding perception, prediction, and planning into one differentiable stack [22, 27]. Dropping LiDAR makes these systems cheaper to deploy, and recent models rival sensor-rich pipelines on open-loop nuScenes [41, 62, 79]. Camera-only E2E driving is no longer a research curiosity but a serious candidate for real vehicles.

A real camera stream is rarely clean. Lenses defocus and

the vehicle vibrates; sensors add noise in low light; weather scatters and occludes the scene; compression and transmission cost quality; frames arrive late or not at all; and image buffers run at reduced refresh to save power flip bits. Each is routine in deployment, and each changes the very pixels the planner consumes. How much of this can an E2E policy absorb before its driving breaks, and which kinds matter most? The answer tells a system designer where to spend the sensing and compute budget, which failures are merely uncomfortable and which are dangerous, and how far the imaging pipeline can be compressed or down-clocked before safety erodes. Existing corruption benchmarks stop one stage too early: they corrupt the input and then measure detection or bird’s-eye-view perception [10, 18, 71]. Perception is not what drives the car. A planner can stay robust where perception is fragile, or fail where perception still looks healthy, because it can lean on ego motion and the navigation command rather than the image [38].

We answer with DriveDegrade (Fig. 1), which injects a recognized corruption taxonomy at five severities, aligned with ImageNet-C and nuScenes-C [18, 71] and extended with four driving-specific contamination families, directly into each model’s image loader, so the same operator at the same severity reaches every architecture and no corrupted copies touch disk. We evaluate open-loop planning on nuScenes with trajectory L2, collision, comfort, and an energy proxy, repeat a slice under NAVSIM’s rule-based PDMS and EPDMS so the findings do not rest on imitation error alone, and anchor the trends closed-loop with two camera-only agents in CARLA across three towns.

Contributions.

- **One injector, fifteen policies, five architecture families.** Planning-level corruption evaluation is not new [28]; what is new is reaching every architecture with the *same* operator at the *same* severity, from a model-agnostic injector inside each model’s own loader (Sec. 3, 4.7, 4.8, 4.9).
- **Per-corruption tolerance thresholds.** An analysis that locates, per corruption, the severity at which planning starts to break, over sixteen families including four

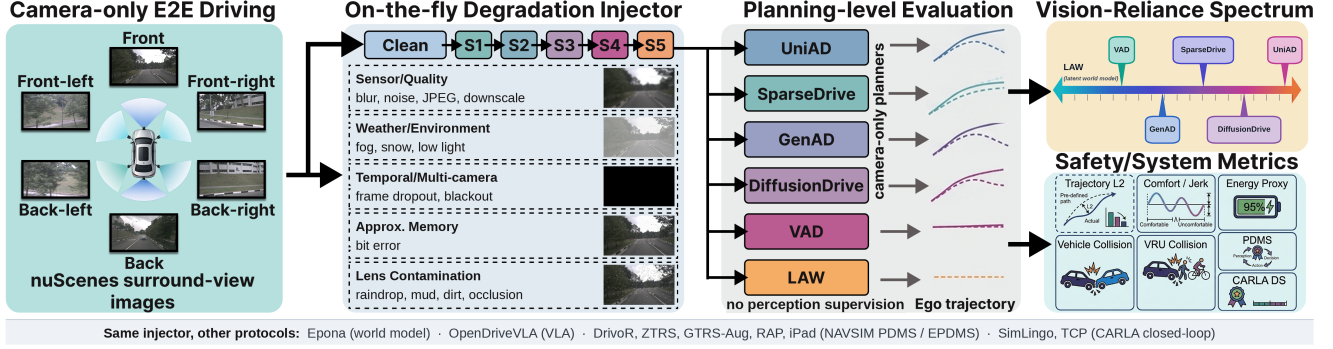


Figure 1. **DriveDegrade overview.** Surround-view camera streams are degraded on the fly inside the image loader over a five-level severity ladder. Six camera-input E2E planners are evaluated at the planning level, then summarized through tolerance curves, a vision-reliance diagnostic separating quality degradation from information removal, and safety/system metrics. The band beneath lists the nine further policies the identical injector reaches under their own protocols. All sixteen corruption families are specified in Supp. B.

driving-specific contamination families whose damage is spatially concentrated rather than global (Sec. 4.4.5).

- **A vision-reliance diagnostic that resolves the flat-curve ambiguity.** Separating quality degradation from information removal turns the degradation curve into an interpretable measure of camera reliance, rather than a robustness score that rewards a planner for ignoring its input (Sec. 4.2).

2. Related Work

Corruption robustness and degraded-driving benchmarks. Common-corruption robustness began with ImageNet-C [18] and has since spanned distribution shift, shape bias, augmentation, backbones, segmentation, and adverse weather [1, 15, 19, 20, 29, 43, 49, 53, 54], then driving perception in three dimensions and LiDAR [10, 30–32, 71]. All share one boundary: they report how much *perception* degrades, not the planned trajectory that moves the vehicle.

Two works probe E2E driving robustness directly. RobustE2E [28] pairs five adversarial attacks and a module-wise attack with 17 natural corruption types at five severities on UniAD, ST-P3, and VAD. Bench2Drive-Robust [78] evaluates four closed-loop agents under deployment perturbations, namely burst frame drop, black-rectangle partial observation, GPS and speed noise, and inference latency, targeting system-level rather than image-level imperfection. Both report results across their severity settings.

Three things do. The prior planning-level study pre-dates the sparse, generative, and diffusion planners. Neither prior work separates quality degradation from information removal, so neither can tell a robust curve from an unread camera (Sec. 4.2). And neither reports per-family knee locations, an approximate-memory fault family, or a memory-interface energy proxy. DriveDegrade is complementary to both: the image-degradation axis Bench2Drive-Robust excludes by design, on architectures RobustE2E does not

cover, with a diagnostic neither provides.

Camera-only end-to-end driving and ego-status shortcuts.

A growing family predicts the ego plan directly from cameras: UniAD couples detection, tracking, mapping, motion, and occupancy before planning [22]; VAD and VADv2 use a vectorized scene [6, 27]; SparseDrive is sparse query-based [62]; GenAD and DiffusionDrive cast planning as generation [41, 79]; LAW plans from a learned latent [35]. Others round out the space, some adding LiDAR [7, 21, 24, 26, 37, 56, 57, 68, 72, 73], all building on camera BEV backbones [23, 34, 36, 42, 44–47, 50, 51, 66, 67]. They differ sharply in how much of the plan depends on the image, yet they are normally compared only by clean accuracy. Open-loop nuScenes planning is largely predictable from ego velocity, acceleration, and the high-level command, so a model can score well while barely using its cameras [38, 75]. This is usually treated as a caveat; we treat it as a measurement, provided the probe distinguishes tolerating a degraded image from disregarding it (Sec. 4.2).

Planning evaluation protocols, robustness methods, and energy.

Open-loop planning is measured on nuScenes and related datasets [2, 4, 14, 60, 69, 74], closed-loop behavior in simulators and planning benchmarks [3, 8, 11, 25, 76]. Neither includes systematic camera-degradation evaluation, and no prior planning-level study spans the modern nuScenes planners at per-family resolution (Table 1). Orthogonal lines improve robustness at training or test time [12, 16, 48, 63, 64] or study domain shift [5, 9, 61, 65]; we instead *measure* tolerance. Our energy proxy is grounded in bus-encoding and approximate-memory work [33, 52, 55, 58, 59].

Table 1. DriveDegrade versus existing robustness and evaluation benchmarks. Perception benchmarks stop at perception; planning benchmarks do not corrupt the image. RobustE2E [28] and Bench2Drive-Robust [78] evaluate planning under perturbation and both report graded severities; Sec. 2 states their scope. *eval. level* is the scored output, OL = open-loop, CL = closed-loop; *img. corr.* is whether the image itself is degraded (adv. = adversarial, sys. = system-level); *plan.* counts evaluated E2E policies, “many” where a benchmark hosts an open leaderboard rather than a fixed set. Ours are six nuScenes planners, a world model, a VLA, five NAVSIM policies, and two CARLA agents, with a camera+LiDAR reference and an image-blind floor as uncounted controls; *v.-r. diag.* records whether a benchmark separates quality degradation from information removal, a property of what is measured, not of how well. “partial” under *img. corr.* means only some corruptions act on the image itself.

benchmark	eval. level	img. corr.	plan.	v.-r. diag.
ImageNet-C [18]	class.	yes	–	no
Dong et al. [10]	3D det.	yes	–	no
RoboBEV [71]	BEV perc.	yes	–	no
Robo3D [31]	3D det./seg.	LiDAR	–	no
RoboDrive [32]	BEV/occ./depth	yes	–	no
Bench2Drive [25]	plan. (CL)	no	many	no
NAVSIM [8]	plan. (OL)	no	many	no
Safety2Drive [76]	plan. (CL)	partial	–	no
RobustE2E [28]	plan. (OL+CL)	yes+adv.	3	no
B2D-Robust [78]	plan. (CL)	sys.-level	4	no
Ours	plan. (OL+CL)	yes	15	yes

3. The DriveDegrade Benchmark

3.1. On-the-fly corruption injection

Corruptions are applied inside the image loader, after decoding and before normalization, so no corrupted copies are written to disk and every model consumes the same operator at the same severity parameter. The injector depends only on numpy/cv2, which is what lets one implementation serve six model environments and two simulators unchanged.

Scope of identical injection. Six families are deterministic functions of the input frame, so every model receives bit-identical pixels. The rest draw a per-frame random realization, and for those what is shared is the operator and its severity, not the individual sample. The released injector now seeds them as blake2b(seed, frame, camera) so that they are bit-identical too. The cells reported here predate that change and used a process-local hash, so their realizations are not reproducible even though the operator and severity are. This affects no reported value, since each cell averages 6019 frames $\times$ six cameras and three seeds shift retention by 0.004 and 0.002 (Sec. 4.4). A reproducibility claim should nonetheless be exact (Supp. C).

3.2. Corruption taxonomy and severity specification

The taxonomy follows ImageNet-C and nuScenes-C/RoboBEV across four groups: sensor/quality, weather/environmental, temporal/multi-camera, and approximate-memory. To these we add a fifth group spe-

cific to a camera carried on a vehicle rather than inherited from generic image robustness, *lens and windshield contamination*, comprising occlusion, mud splash, raindrop, and lens dirt. These four differ from blur or noise in kind rather than degree. Their damage is spatially concentrated, so part of the frame stays pristine while the rest carries no information, and no global denoising recovers what an opaque deposit removed. Occlusion is defined as a black rectangular mask matching the partial-observation perturbation of closed-loop deployment benchmarks [78], so the two ladders are directly comparable at the overlap.

Every family is defined by a single physical parameter, and Supp. B lists its value at each of S1–S5 for all sixteen evaluated families, emitted directly from the released injector so that specification and code cannot diverge. The ladder is therefore a specification rather than an ordinal label: a threshold at $s_{10}^* = 2.5$ for gaussian blur reads back as a defocus σ of roughly 1.5 px. Endpoints were fixed before running any model by one rule: S1 is a condition a deployed camera meets routinely, S5 is where a human would call the view unusable, bracketing the knee rather than centering it. Snow and camera blackout have no natural scalar: their parameter is an ordinal, and blackout ramps over camera identity rather than magnitude. Six further families the injector defines are excluded for measured or structural reasons (Supp. B).

3.3. Models

We evaluate six camera-input E2E planners spanning architectures: UniAD (full-stack) [22], VAD (vectorized) [27], SparseDrive (sparse) [62], GenAD (generative) [79], DiffusionDrive (diffusion) [41], and LAW (latent world model) [35]. LAW is trained without perception *supervision*, not blind: it consumes the surround images and responds when they are removed (Sec. 4.2). All models run inference on the full nuScenes validation set (6019 key-frames, six cameras); trajectory L2 is averaged per each model’s official protocol, over the 5119-frame valid-planning subset for GenAD and DiffusionDrive and the full 6019 for the others. Those protocols differ, but the difference is inert for retention: recomputing UniAD on the 5119-frame subset reproduces every retention we report to within 0.0024, two orders of magnitude below the effects we measure (Supp. A). Absolute L2 remains not strictly comparable across the two groups (Sec. 4.3).

3.4. Metrics and tolerance threshold

DriveDegrade evaluates curves rather than a single score. We use retention rather than raw degraded L2 because the planners have different clean baselines and, in two cases, different official valid-planning subsets; normalizing each degraded run by its paired clean run makes the primary comparison the *shape* of degradation. Absolute metrics are

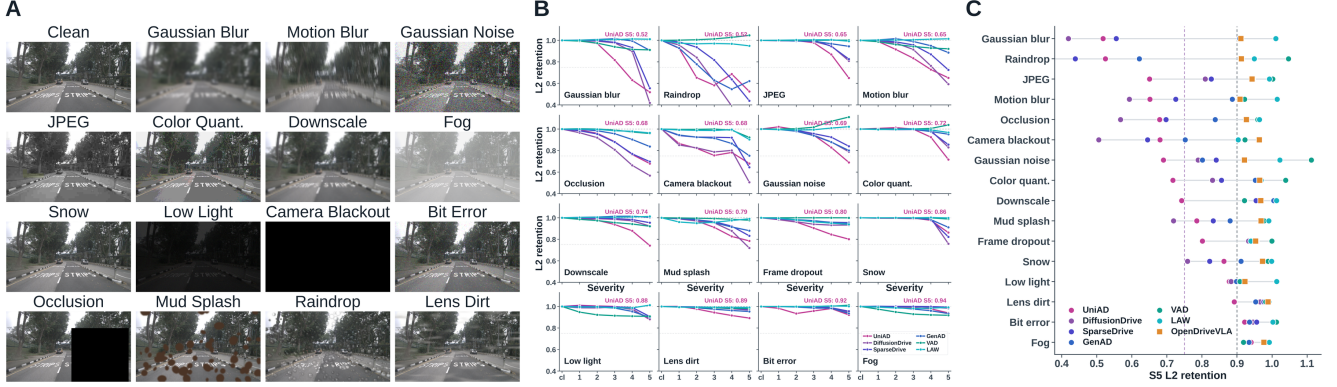


Figure 2. **Per-corruption image-degradation tolerance.** (A) The sixteen families on one nuScenes front-camera frame at S4, clean for reference; the bottom row is the four driving-specific families (Sec. 4.5). Frame dropout has no single-frame rendering. (B) L2-retention curves for all six planners, ordered by UniAD fragility. (C) S5 retention for every corruption and planner. Blur, JPEG, and noise damage planning most; weather and bit-error are tolerated. OpenDriveVLA (squares) sits in $[0.909, 0.989]$ on all sixteen.

still reported where interpretable, especially in the CARLA anchor. Per (model, corruption, severity) we report trajectory L2 at 1/2/3 s, vehicle and VRU collision, comfort, and an energy proxy. We define relative retention $r(s) = M(\text{clean})/M(s)$, where $M(s)$ is the model’s mean L2 at severity s and $M(\text{clean})$ its clean-run value; for the reward-valued PDMS and EPDMS (Sec. 4.6, 4.7) it is the reciprocal $M(s)/M(\text{clean})$, so higher is always better and 1.000 always means no measurable damage. We define the tolerance thresholds s_{10}^*, s_{25}^* as the severity at which r first drops below 0.90, 0.75. Open-loop evaluation is deterministic over a fixed per-model validation split, so each per-cell metric is a population mean rather than a sample estimate: a frame-level bootstrap puts 95% confidence intervals below 0.01 in retention. We therefore omit per-point error bars open-loop and report them only for the stochastic CARLA anchor. Supp. H gives the protocol for adding a planner.

4. Results

4.1. Tolerance thresholds

On UniAD (full nuScenes val, clean L2 = 0.94 m), mild degradation barely matters and each graded image-quality corruption shows a clear knee (Fig. 2B, Table 2). Among these image-quality families, blur and noise are most fragile ($s_{10}^* \approx 2.2$ – 2.5 for motion/gaussian blur); weather and bit-error are tolerated across the whole range. Camera blackout is a separate sensor-loss class rather than a graded degradation: it removes whole cameras, so its $s_{10}^* = 1.0$ (the first severity that drops a camera) is not comparable to the gradual thresholds and we report it apart. Figure 2C ranks the families by S5 retention across all six models.

4.2. Cross-model perception reliance

Perception reliance differs sharply across architectures (Fig. 3, Table 3). Averaged over the twelve corruptions at

Table 2. **Tolerance thresholds s_{10}^* for all six planners** on full nuScenes val: the severity at which L2 retention first falls below 0.90, linearly interpolated. “–” means the threshold is never reached. Rows are ordered by UniAD fragility. The ordering is consistent: whenever a corruption breaks a more camera-reliant planner it breaks it earlier, and VAD and LAW cross no threshold at all, the flat end that Sec. 4.2 disambiguates.

corruption	UniAD	DiffDr	SpDr	GenAD	VAD	LAW
<i>graded image degradation</i>						
motion blur	2.19	3.10	3.66	4.79	–	–
gaussian blur	2.47	3.96	4.09	–	–	–
gaussian noise	3.38	3.69	4.29	3.65	–	–
jpeg	3.72	4.35	4.38	–	–	–
downscale	3.64	–	–	–	–	–
color quant.	4.12	4.55	4.63	–	–	–
frame dropout	3.07	–	–	–	–	–
snow	4.67	4.41	4.52	–	–	–
low light	4.77	4.81	–	4.97	–	–
fog	–	–	–	–	–	–
bit error	–	–	–	–	–	–
<i>driving-specific contamination</i>						
raindrop	1.10	1.49	2.31	1.19	–	–
occlusion	2.78	2.18	2.73	4.05	–	–
mud splash	3.13	3.71	4.25	4.42	–	–
lens dirt	4.68	–	–	–	–	–
<i>sensor-loss stress test</i>						
camera blackout	1.00	1.00	4.07	3.31	–	–

Camera blackout removes whole cameras rather than grading an image parameter, so its threshold is not comparable to the graded families; 1.00 means the mildest setting already crosses. DiffDr = DiffusionDrive, SpDr = SparseDrive. The stricter s_{25}^* thresholds are in Supp. B.

S5, mean L2 retention places the full-stack and diffusion planners at the camera-reliant end, the sparse and generative ones in the middle, and the vectorized and latent-world-model ones barely moving. The ordering holds per corruption and is steepest under blur, where DiffusionDrive and UniAD keep only 0.42 and 0.52 of clean L2 (Fig. 2B).

Quality versus removal. That spectrum is ambiguous at its flat end, which is exactly where one would otherwise de-

clare a winner. We split the corruption set by what it does to the information in the image: quality families leave the scene present but worse, removal families delete it. That supports an asymmetric test needing no extra runs. A planner that reads its camera must lose accuracy when information is removed, whatever it does under quality loss, and one flat on both axes is not using the camera.

Table 3 applies it. None of the six is flat on both axes, so none ignores the camera outright, though Sec. 4.8 adds a released VLA that very nearly is. UniAD, DiffusionDrive, SparseDrive, and GenAD degrade on both. VAD and LAW are nearly immune to quality (0.97 and 1.00) yet still lose 7.7% and 9.7% when the front camera is removed, and LAW declines monotonically under frame dropout (0.996 $\rightarrow$ 0.940). Their flatness is insensitivity to image *quality*, not blindness: they use the coarse presence of visual evidence but not its detail. That is the ego-status shortcut [38] stated precisely, and a weaker claim than a one-axis reading would license. The gap between the axes is itself informative: largest for planners that tolerate bad pixels but not missing ones, smallest where dependence on fine detail means degradation destroys as effectively as deletion.

4.3. Absolute performance and rank reversal

Retention normalizes away clean accuracy, so Table 3 also reports clean L2 and we release every absolute cell. Two things follow that retention cannot express. First, degradation reorders the leaderboard: GenAD beats VAD on clean data (0.908 against 0.969 m) and loses to it under gaussian noise S5 (1.133 against 0.871 m), so a planner that is not state of the art clean can be the better choice under degradation. Second, the planner trained without perception supervision has the lowest absolute L2 in *every* condition

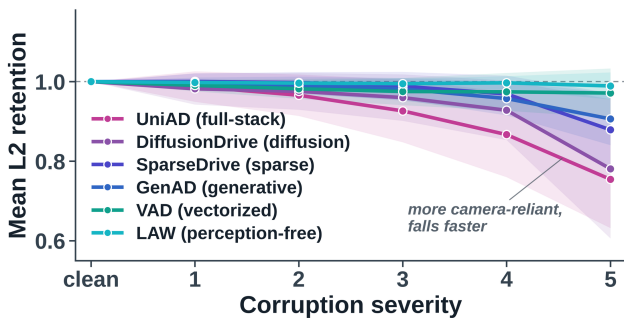


Figure 3. Mean L2 retention vs. severity. The slope indexes camera reliance: the full-stack and diffusion planners fall fastest, the vectorized and latent-world-model planners stay flat, an insensitivity to image quality that Table 3 separates from disregard of the camera. The six solid curves average the twelve core corruptions; OpenDriveVLA (dashed) has full ladders for four of them, and the ego-only MLP (dotted) is 1.000 everywhere by construction. On those four families the six planners run 0.625 to 0.980 at S5, so OpenDriveVLA’s 0.932 sits between VAD and LAW.

Table 3. **Vision reliance on two axes.** S5 L2 retention under corruptions that *degrade* the image versus those that *remove* it. A planner ignoring its camera would sit at 1.00 in both columns; none of the six does. VAD and LAW are quality-immune yet still lose accuracy when a camera is removed, so their flat curves measure insensitivity to quality, not blindness. The VLA is the only entry with a *negative* gap. “gap” is quality minus blackout retention; bold marks the largest.

model	clean L2 (m)	S5 retention				gap
		mean	qual.	black.	drop.	
UniAD	0.942	0.75	0.739	0.680	0.801	0.059
DiffusionDrive	1.065	0.78	0.777	0.505	0.933	0.271
SparseDrive	0.608	0.84	0.831	0.645	0.955	0.186
GenAD	0.908	0.91	0.915	0.752	0.948	0.163
VAD	0.969	0.97	0.969	0.923	0.999	0.047
LAW	0.524	0.99	1.003	0.903	0.940	0.099
OpenDriveVLA	0.677	0.95	0.943	0.964	0.953	−0.021
ego-only MLP	0.731	1.00	1.000	1.000	1.000	0.000

“mean” averages all twelve core S5 corruptions; “quality” the nine that degrade rather than remove (gaussian/motion blur, gaussian noise, JPEG, color quantization, downscale, fog, snow, low light). Retention is clean L2 over degraded L2. Per-family values for all sixteen families are in Fig. 2C. The last row is our ego-status-only regressor (no image input, Sec. 5), 1.000 everywhere by construction. OpenDriveVLA’s mean covers the eleven core families it reports at S5.

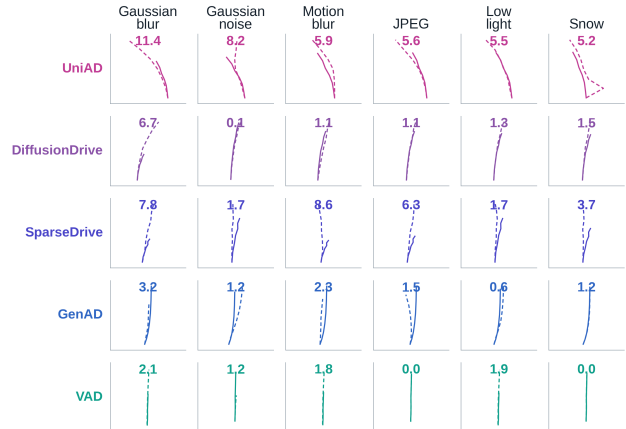


Figure 4. **Cross-method qualitative.** One nuScenes scene’s planned ego trajectory for five planners (rows) under the six most disruptive families at S5 (columns, ordered by UniAD’s endpoint shift): clean (solid) vs. corrupted (dashed), with the 3 s endpoint shift in metres. Drift scales with camera reliance: UniAD and SparseDrive swing widely under blur, noise, and JPEG while VAD stays almost straight. LAW is omitted because its stored outputs are aggregate metrics, not trajectory points.

tested: LAW is 0.524 m clean and never exceeds 0.581 m anywhere. Our ego-status-only floor makes the same point from below, reaching 0.731 m with no image at all, ahead of four of the five camera-supervised planners. On open-loop nuScenes L2 no amount of visual supervision buys accuracy over a model leaning on ego state [38, 75]. Absolute L2 therefore cannot certify that a planner uses its camera well, which is why the degradation *slope* and the removal axis carry the diagnostic weight. Scaling every model against

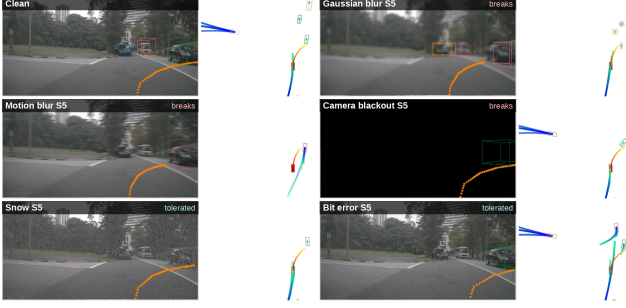


Figure 5. **Image evidence versus plan stability (UniAD, one scene).** Each cell pairs the front camera with UniAD’s planned trajectory under one S5 corruption: clean, the three families that break the plan, and two that are tolerated. Blackout removes the image entirely yet the plan bends less than under blur, which is the asymmetry Sec. 4.2 measures. The full twelve-corruption montage with all six surround views is in Supp. I.

one reference, as RoboBEV [71] does for BEV detectors, assumes a shared protocol. That holds for a single detection benchmark but not for planners shipping different evaluators, horizons, and subsets, where the reference model’s protocol becomes everyone’s yardstick. Releasing every absolute cell lets any baseline-relative score be recomputed.

4.4. Secondary metrics and checks

The planned path bends in proportion to camera reliance (Fig. 4, Fig. 5): under gaussian blur S5 UniAD shifts its 3 s endpoint by 11.4 m and DiffusionDrive by 6.7 m, while VAD barely moves (2.1 m). Comfort and vehicle collision follow the same ordering, UniAD’s jerk rising from 1.92 to 2.69 m/s³ and its 3 s collision rate from 0.65% to 1.14% (Fig. 6A). VRU collision does not: at 0.02–0.25% it is a rare-event metric, flat across severity for every model, because on a replayed scene the drifted plan does not preferentially strike a VRU. The memory-interface energy proxy (bit-1 density relative to clean, Fig. 6C) estimates the cost of moving and storing image bytes [33, 52, 58], not system power: on an automotive SoC the dominant term is network computation, which degradation leaves unchanged. Within that scope, low light and color quantization cut write energy while retaining 0.88 and 0.72. Finally, the stochastic families are seed-independent: three fresh seeds shift retention by 0.004 and 0.002 on the two most stochastic cells, against per-family effects of 0.1–0.5 (Supp. C).

4.5. Driving-specific contamination

The four contamination families (Fig. 2A, bottom row) evaluate a failure mode the generic taxonomy cannot represent: spatially concentrated damage. They reproduce the reliance spectrum of the generic taxonomy but reorder its fragile end (Table 2, middle block). Raindrop is the most damaging family in the benchmark for the camera-reliant planners, crossing s_{10}^* between 1.10 and 2.31 where no generic family

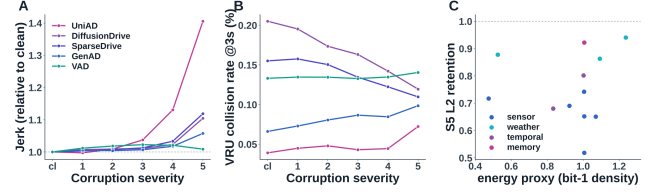


Figure 6. **Secondary metrics across planners.** (A) Comfort: jerk relative to each planner’s clean baseline, mean over the twelve corruptions. UniAD degrades most ($\sim 1.4\times$ at S5) while VAD barely changes, mirroring the L2 spectrum. Absolute jerk is not comparable across planners, so (A) is normalized per planner. (B) VRU collision rate: small (~ 0.02 – 0.25%) and roughly flat, a rare-event metric. (C) Energy proxy (bit-1 density) vs. S5 retention by corruption group (UniAD). “cl” = clean.

crosses before 2.19. GenAD is the sharpest case: it crosses only four thresholds among the twelve generic families, yet raindrop breaks it barely past the mildest setting, so local refractive distortion defeats a planner that global degradation hardly moves. Occlusion, the pure removal member, lands in the same band as gaussian blur even though half of every frame stays pristine at S5, so these planners integrate evidence across the full frame rather than from a region occlusion could spare. Epona reproduces the raindrop result on its own protocol (Supp. F).

4.6. Rule-based scoring: PDMS and EPDMS

Open-loop L2 measures agreement with one recorded future. NAVSIM’s PDMS instead rewards collision avoidance, drivable-area compliance, and time-to-collision [8], and the conclusions hold under it (Table 4). On navtest, 12,147 scenes per cell, TransFuser’s PDMS falls under blur and further under front-camera blackout, with every safety sub-score declining alongside it. Its retention is far milder than the camera-only planners’ because LiDAR is untouched by image corruption, which is why the main benchmark is camera-only: a sensor-rich agent understates the effect being measured. Isolating the backbone from the modality, camera-only DrivoR on a DINOv2 ViT-S backbone matches those retentions (0.956–0.962) with no second modality, so a foundation-model backbone alone recovers what the ImageNet-pretrained camera-only planners lack. The recovery is not unconditional: front-camera blackout collapses DrivoR from 0.937 to 0.531. Sub-scores in Supp. D.

4.7. Four NAVSIM planners on the same slice

Four NAVSIM-era planners run over the same nine-cell slice, thirty-six evaluations (Table 5). They do not share one protocol and we do not force one on them: GTRS-Aug [40] and ZTRS [39] publish on navhard.two_stage under EPDMS, RAP [13] and iPad [17] on navtest under PDMS, so the absolute columns must not be pooled or ranked. Retention divides each model by its own clean

Table 4. **PDMS under degradation** (NAVSIM *navtest*, 12,147 scenes per cell). TransFuser fuses camera with LiDAR; DrivoR is camera-only on a DINOv2 ViT-S backbone. NC = no collision, DAC = drivable-area compliance, TTC = time-to-collision, EP = ego progress. Degradation lowers PDMS and every safety sub-score, so the effect is not an artifact of imitation L2. S3 blackout removes only rear cameras and moves neither agent; front-camera blackout is the one corruption the backbone cannot absorb.

condition	PDMS	ret.	NC	DAC	TTC	EP
<i>TransFuser (camera+LiDAR)</i>						
clean	0.834	1.000	0.978	0.921	0.928	0.786
jpeg S5	0.807	0.968	0.968	0.906	0.903	0.766
motion blur S5	0.805	0.965	0.968	0.911	0.905	0.755
gaussian blur S5	0.797	0.956	0.966	0.905	0.905	0.741
blackout S3 (rear)	0.834	1.000	0.978	0.921	0.928	0.786
blackout S5 (front)	0.753	0.903	0.933	0.886	0.845	0.724
<i>DrivoR (camera-only, DINOv2 ViT-S)</i>						
clean	0.937	1.000	0.990	0.989	0.967	0.899
jpeg S5	0.896	0.956	0.985	0.962	0.953	0.853
motion blur S5	0.901	0.962	0.980	0.977	0.941	0.860
gaussian blur S5	0.900	0.961	0.979	0.972	0.925	0.880
blackout S3 (rear)	0.937	1.001	0.990	0.989	0.967	0.900
blackout S5 (front)	0.531	0.567	0.832	0.697	0.757	0.479

run, which is what makes the shapes comparable. Every clean cell reproduces its published figure before any corrupted cell is read: ZTRS 45.45 against 45.5 and RAP 93.80 against 93.8 exactly, iPad within a tenth, GTRS-Aug 0.8 EPDMS high.

Null control. Blackout S3 darkens the three rear cameras, which none of these planners reads, so each reproduces its clean score bit for bit, identical to fourteen decimal places. That is four confirmations, in four separately patched devkits, that the injector is inside the loader and doing what the specification says, at no extra evaluation cost.

Quality-axis retention is mild for all four and almost absent for RAP (0.992–1.001 on a DINOv3 backbone), yet deleting the front camera costs RAP half its score and GTRS two thirds. A foundation-model backbone therefore buys near-immunity to degraded pixels, not to missing ones, which bounds Sec. 4.6. Porting details in Supp. E.

4.8. A world model and a vision-language-action planner

Two families proposed as successors to the imitation-trained design would be invisible to a benchmark measuring only the first, so we ran *Epona* [77], an autoregressive diffusion world model, and *OpenDriveVLA-0.5B* [80]. Neither buys robustness: *Epona* shows the camera-reliant profile and reproduces our raindrop finding independently. *OpenDriveVLA* is the informative case: its clean L2 reproduces the published 0.35 m, yet no cell in a forty-eight-cell sweep moves it, every retention falling in [0.909, 0.997], and the front camera going dark costs 3.6%, so it is flat on both axes and alone in having a *negative* gap (Table 3). Zeroing

Table 5. **Four NAVSIM planners**, nine cells each. The clean row is absolute; the rest are retention (corrupted/clean). GTRS and ZTRS score EPDMS on *navhardtwo-stage*, RAP and iPad PDMS on *navtest*, so absolute values are *not* comparable across the rule. Blackout S3 darkens the three rear cameras, which none of them reads: all four reproduce clean bit for bit, to fourteen decimals, a null control on the injector. Blackout S5 costs every planner a quarter to two thirds of its score.

condition	EPDMS, <i>navhard</i>		PDMS, <i>navtest</i>	
	ZTRS	GTRS	RAP	iPad
clean (abs.)	0.4545	0.4294	0.9380	0.9179
gauss. blur S3	0.9577	1.0017	0.9998	0.9901
gauss. blur S5	0.8815	0.9390	0.9984	0.9474
motion blur S3	0.9432	0.9630	1.0011	0.9900
motion blur S5	0.9269	0.9031	0.9979	0.9608
jpeg S3	0.9628	0.9608	0.9987	0.9948
jpeg S5	0.7874	0.8618	0.9915	0.9627
blackout S3 (rear)	1.0000	1.0000	1.0000	1.0000
blackout S5 (front)	0.4624	0.3513	0.4939	0.7727

all six views on the full validation set still leaves retention 0.885: blinding it entirely costs 11.5%, where UniAD loses 32% from the front camera alone. Its open-loop accuracy is carried by ego status and the command: the AD-MLP and BEV-Planner critique [38, 75] inside a VLA, found by a corruption ladder rather than by ablating an input. Details in Supp. F.

4.9. Closed-loop anchor in CARLA

Camera-only agents anchor the trends under feedback, since a fusion planner would cover a degraded camera with LiDAR. With all twelve families injected into SimLingo’s camera stream on Town10, front-camera blackout at S5 reduces mean Driving Score from 100 to **19.4** while image-quality conditions stay above 81 (Table 6). Complete-case averaging is optimistic exactly where the benchmark bites, so we bound the unscored runs rather than drop them: assigning them any DS in [0, 100] puts blackout S5 in [15.6, 35.6] and the widest image-quality interval in [53.5, 93.5], so the availability-versus-quality separation survives any assignment (Fig. 7). Rankings *within* the image-quality group do not. Repeating the slice on Town03 and Town05 reproduces the ordering in both, with collision counts an order of magnitude above any other cell.

Second agent and null control. One agent cannot carry a generalization, so we ran the identical Town05 slice with TCP [70]. The slice contains a *free null control*: blackout S3 names the three rear views and neither agent has one, so it changes no pixel and measures how far a driving score wanders on rerun with the input untouched. SimLingo’s null does not move, all eight runs reproducing clean exactly, so its blackout-S5 collapse is unambiguous. TCP’s moves four of twelve runs at mean -12.8 DS, as large as any effect we measure on it, so *no TCP cell separates from its own null* and we report that instead of per-cell means. Blackout

Table 6. **CARLA closed-loop Driving Score.** The top block is the seven-cell slice on three towns and, last column, a second agent; the middle block the remaining Town10 corruptions (S3/S5); the bottom the contamination families on Town03. Blackout S3 is the null control of Sec. 4.9: SimLingo’s does not move, TCP’s moves four of twelve runs. DS conditions on completed scored runs.

condition	SimLingo			TCP
	Town10	Town03	Town05	Town05
clean	100.0	92.5	90.0	68.8
gaussian blur S3	100.0	92.5	95.0	68.7
gaussian blur S5	90.2	72.0	85.0	55.1
motion blur S3	96.4	96.2	95.0	71.7
motion blur S5	94.7	92.5	87.0	68.2
blackout S3 (null)	100.0	92.5	90.0	57.8
blackout S5 (front)	19.4	39.8	36.3	60.9
<i>remaining Town10 corruptions, S3/S5</i>				
fog	100.0/100.0	—	—	—
bit error	100.0/100.0	—	—	—
frame dropout	100.0/100.0	—	—	—
jpeg	92.0/91.1	—	—	—
snow	100.0/89.2	—	—	—
gaussian noise	100.0/89.1	—	—	—
low light	100.0/88.6	—	—	—
color quant	96.0/88.6	—	—	—
downscale	100.0/81.6	—	—	—
<i>contamination families, Town03, S3/S5</i>				
lens dirt	—	92.5/87.5	—	—
mud splash	—	92.5/64.5	—	—
raindrop	—	89.9/65.6	—	—
occlusion	—	84.3/64.3	—	—

is also not the same treatment for the two, since SimLingo goes blind while TCP loses the center third of a panorama.

Contamination families. Run closed-loop on Town03, the open-loop ordering survives at the ends and collapses in the middle: lens dirt stays mildest while occlusion, mud splash, and raindrop converge at S5 to within 1.3 DS of each other, where open loop separates them by more than a severity step. What differs is the *distribution* of the loss. Occlusion spreads it across five of eight runs; raindrop and mud splash leave six runs untouched at 100 and annihilate both seeds of the one route already below par. Deleting part of the view costs a little almost everywhere, whereas refractive distortion costs nothing until the route demands what it hides, and a per-cell mean reports the two as the same number. Six runs end on the blocked-actor criterion, all in severe cells; counting only completed runs would report raindrop S5 at 86.7 rather than 65.6. Route traces and per-run scores are in Supp. G.

5. Evaluation Scope and Limitations

DriveDegrade supports comparative claims under controlled conditions: model, split, evaluator, and operator are fixed, and only the camera stream is degraded. It measures tolerance thresholds and relative vision reliance, not deployment safety. Because open-loop nuScenes rewards ego-status shortcuts, a flat curve is not by itself evidence

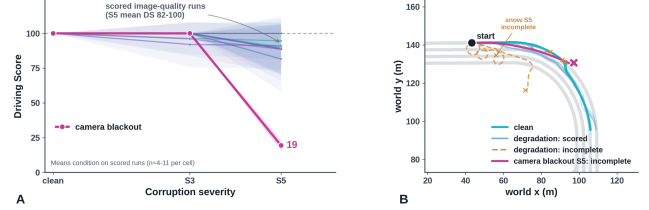


Figure 7. **CARLA closed-loop evidence and run status.** (A) Complete-case DS on Town10; bands are ± 1 std over scored runs. (B) Route-148 traces: solid blue is scored degradation, dashed orange incomplete, magenta incomplete blackout S5. The snow-S5 loop is excluded from the DS mean, which the identification bounds account for.

of quality; the quality-versus-removal split of Sec. 4.2 separates robustness from indifference. None of the six planners is fully non-visual, so we pin the blind floor with an ego-status-only regressor in the style of AD-MLP [75]: a planner whose blackout-S5 accuracy stays above that floor is extracting value from its remaining cameras.

Four limitations follow. Open-loop nuScenes measures replayed planning error, so collision and comfort miss feedback compounding. The closed-loop anchor spans two agents and three towns but remains one simulator with few routes; the identification bounds keep the blackout separation robust to unscored outcomes but do not justify fine-grained ranking among image-quality conditions, and TCP’s null moves as much as its largest effect. The corruptions are controlled synthetic stressors, not all real sensor physics, and the energy figure is an analytical proxy rather than measured watts.

6. Conclusion

DriveDegrade turns camera degradation from a perception-only robustness question into a planning-level tolerance measurement. Across sixteen corruption families, with one injector reaching fifteen policies, it locates clear breakpoints for blur, JPEG, noise, and camera loss while weather and bit-error are often tolerated. The same curves reveal vision reliance that clean-set scores hide, and the CARLA anchor confirms camera loss remains load-bearing under feedback. **Artifact.** The injector, hooks, configs, scripts, and per-cell CSVs will be released at <https://github.com/UGA-MOBILITY-LAB/DriveDegrade>, MIT licensed and with no raw sensor data. Every cell is traceable to a checkpoint, split, corruption, severity, and seed, and every table and figure regenerates from them on CPU. We version the specification and treat operator changes as major versions, so published numbers stay valid (Supp. H).

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